

Project 3: Vector-Graph Cross-Linking: Powerful insights by combining semantic and relational search

Clinical Intelligence System with Hybrid Database Architecture

Project Overview

This mini-project demonstrates how **cross-linking Vector and Graph databases** creates powerful clinical insights by combining:

- **Semantic Search** (Vector DB) - Find similar clinical cases, symptoms, and treatments
- **Medical Knowledge Graph** (Graph DB) - Navigate relationships between diseases, drugs, symptoms, and patients
- **Cross-Linking** - Reference graph entities from clinical text chunks and vice versa

The Healthcare Problem

Scenario: A physician sees a patient with complex symptoms. They need to:

1. Find similar cases based on symptom descriptions (semantic search)
2. Understand drug interactions and contraindications (relationship traversal)
3. Discover treatment patterns from patients with similar conditions
4. Navigate from a clinical note to the patient's full medical graph

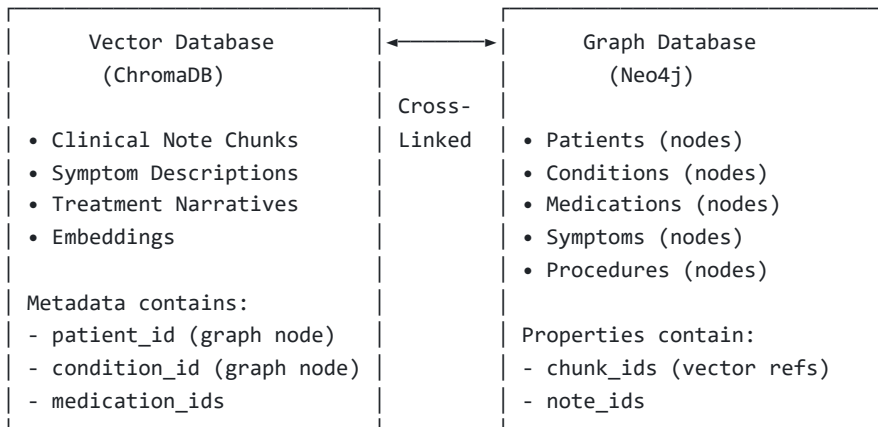
Traditional Approaches Fall Short:

- Keyword search misses semantically similar cases
- Structured databases can't search free-text clinical notes effectively
- Graph databases alone can't find "similar" unstructured content
- Vector databases alone miss critical relationships (drug interactions, comorbidities)

Cross-Linked Solution:

- Semantic search finds relevant clinical narratives
- Instant traversal to related entities (patient, conditions, medications)
- Navigate the medical knowledge graph from any search result
- Unified view of structured and unstructured clinical data

Architecture



Use Case: Clinical Decision Support

Example Workflow

1. Doctor's Query: "Find cases with chronic fatigue, joint pain, and elevated inflammatory markers"

2. Hybrid System Response:

- Vector DB finds semantically similar clinical notes
- Each note links to patient nodes in the graph
- Graph reveals:
 - Diagnosed conditions (Lupus, Rheumatoid Arthritis, Fibromyalgia)
 - Current medications and known interactions
 - Comorbidities and contraindications
 - Treatment outcomes

3. Clinical Insights:

- Similar cases across the patient population
- Treatment patterns that worked
- Drug interactions to avoid
- Specialist referral patterns

Quick Start

Installation

1. Install Python dependencies

```
pip install chromadb neo4j sentence-transformers pandas numpy
```

3. Run the project

```
python 01_setup_healthcare_dbs.py
```

```
# Initialize databases with sample data
```

```
python 02_clinical_queries.py
```

```
# Run example clinical queries
```

```
python 03_demonstrate_value.py
```

```
# See the power of cross-linking
```

Project Structure

```
citius-healthcare-vectorgraph/
├── README.md                # This file
├── clinical_data.json       # Sample patient data
├── 01_setup_healthcare_dbs.py # Load sample data into both DBs
├── 02_clinical_queries.py   # Example hybrid queries
├── 03_demonstrate_value.py   # Demonstrate cross-linking benefits
├── 04_interactive_demo.py    # Interactive CLI demo
├── utils/
│   ├── vector_db_manager.py  # Vector DB operations
│   ├── graph_db_manager.py   # Graph DB operations
│   └── clinical_search.py    # Cross-linked search functions
```

Learning Objectives

1. Understand Database Strengths

- **Vector DB:** Semantic similarity in clinical narratives
- **Graph DB:** Medical relationships and contraindications
- **Together:** Comprehensive clinical intelligence

2. Cross-Linking Patterns

- Store graph node IDs in vector metadata
- Store vector chunk IDs in graph node properties
- Bidirectional navigation between systems

3. Real Healthcare Applications

- Clinical decision support
- Similar case retrieval
- Drug interaction checking
- Patient journey analysis
- Research cohort discovery

Example Queries You'll Build

Query 1: Symptom-Based Case Finding

```
"Find patients with unexplained weight loss and night sweats"
→ Vector search finds similar symptom descriptions
→ Link to patient graphs
→ Reveal diagnoses: Lymphoma, Tuberculosis, Hyperthyroidism
```

Query 2: Drug Interaction Discovery

"Show me patients on warfarin who were prescribed antibiotics"

- Graph traversal finds medication combinations
- Link to clinical notes documenting outcomes
- Identify interaction patterns and adverse events

Query 3: Treatment Pattern Analysis

"For Type 2 Diabetes with CKD, what treatments have best outcomes?"

- Graph finds patients with both conditions
- Vector search retrieves treatment notes
- Analyze outcomes and medication effectiveness

Query 4: Comorbidity Navigation

"Find similar cases to this patient with multiple chronic conditions"

- Vector search on clinical presentation
- Graph reveals comorbidity networks
- Discover hidden associations and risk factors

Expected Outcomes

After completing this project, you'll see:

✅ **Better Clinical Search:** Semantic understanding of medical terminology ✅ **Relationship Discovery:** Hidden connections in patient data ✅ **Safety Checks:** Drug interactions and contraindications ✅ **Evidence-Based Insights:** Treatment patterns from real cases ✅ **Unified View:** Structured + unstructured data together

ample Data Included

- **8 Patients** with realistic clinical profiles
- **12 Medical Conditions** (Diabetes, Hypertension, Cancer, etc.)
- **15 Medications** with interaction data
- **20+ Clinical Notes** with rich narratives
- **Symptoms, Procedures, Labs** interconnected in the graph

Key Concepts Demonstrated

1. Bidirectional Cross-Linking

```
# In Vector DB metadata:
{
  "patient_id": "PAT001",           # Link to graph node
  "condition_ids": ["COND001"],     # Links to condition nodes
  "chunk_type": "clinical_note"
}

# In Graph DB node properties:
Patient {
  id: "PAT001",
  chunk_ids: ["chunk_001", "chunk_002"] # Links to vector chunks
}
```

2. Semantic + Structural Search

Combine the power of:

- Finding semantically similar text
- Traversing medical knowledge relationships
- Filtering by structured attributes

3. Clinical Safety

- Check drug interactions via graph traversal
- Find contraindications based on conditions
- Identify risk factors through relationship patterns

Healthcare Compliance Notes

This is a **demonstration project with synthetic data only**.

For production use in healthcare:

- Implement proper HIPAA compliance
- Add audit logging
- Implement access controls
- Encrypt PHI (Protected Health Information)
- Follow data retention policies

Notes for Workshop Participants

This project includes:

-  **Extensive comments** explaining every step
-  **Print statements** to see what's happening
-  **Error handling** with helpful messages

-  **Modular code** that's easy to understand
-  **Progressive complexity** from simple to advanced
-  **Interactive demo** to explore on your own

Interactive Demo

Run the interactive demo to explore queries yourself:

```
python 04_interactive_demo.py
```

Try queries like:

- "chest pain and shortness of breath"
- "diabetes medication with kidney problems"
- "patients with multiple chronic conditions"

Next Steps

After completing this project:

1. Add your own clinical scenarios
2. Implement more complex graph patterns (3+ hop traversals)
3. Add temporal analysis (disease progression over time)
4. Build a simple web UI
5. Integrate with HL7 FHIR data standards